CRASH COURSE #2 · THURSDAY 22 OCTOBER 2026 · 17:00 → 19:00 · ON ZOOM

Vibe coding & AI-assisted development

Direct the agent. Check the result.

Tonight you open your challenge repository, put an AI agent to work in it, and learn to catch it when it is confidently wrong.

FacMed Hack #1 · Digital Phenotyping of Mental Health · Fri 6 – Sat 7 November 2026 · Campus Biotech · in English

The road to 6 November

Thu 8 Oct · B1 · Digital phenotyping & the Swiss data rules
Today · B2 · Vibe coding & AI-assisted development
Thu 29 Oct · B3 · Grant writing for pop-up projects, with the RGO
Fri 6 – Sat 7 Nov · the hackathon, 32 hours

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4 min

BEFORE ANYTHING ELSE · 90 SECONDS · ONE OR TWO LINES IN THE ZOOM CHAT

Where do you imagine vibe coding and agentic coding will change science? Name one place, and what changes there.

Your own field or any other. A method, an instrument, a job in the lab, a stage of publishing: be specific.

> *"What would you do if you had all the models of the brain you wanted?"*
>
> Grace Lindsay, NYU, responding to this week's pre-read (DePasquale, *The Transmitter*, June 2026)

8 min

SIX QUESTIONS ABOUT WHERE THIS IS GOING

Five have no right answer. One does.

How

One question per slide, about a minute each. A Zoom poll pops up with each one. Anonymous. Answer with what you actually expect for your field.

Why

The split in the room is the material for tonight. We keep the results, set them against what researchers in the field are saying, and come back to them before 19:00.

No poll on your screen? Type your letter in the chat.

QUESTION 1 OF 6 · WHERE FIRST

Where will agentic coding change research practice most in the next five years? *Pick up to two.*

- **A** Analysing data and building models
- **B** Designing and running experiments: automated labs
- **C** Reading and synthesising the literature
- **D** Writing papers and grants
- **E** Peer review and replication

QUESTION 2 OF 6 · THE SCARCE SKILL

In 2031, the scarcest skill in a neuroscience lab will be...

- **A** Programming
- **B** Statistics and methods
- **C** Choosing the question worth asking
- **D** Judging whether a result is true
- **E** Collecting good data from real people

QUESTION 3 OF 6 · DEPTH

This week's pre-read warns that AI coding could make theoretical neuroscience *"prolific but shallow"*. For your own field, you think...

- **A** It is already happening
- **B** It will happen unless incentives change
- **C** The opposite: faster coding frees time to think
- **D** Too early to say

B. DePasquale, "Models at the speed of thought: How AI coding is reshaping theoretical neuroscience", *The Transmitter*, 16 June 2026.

QUESTION 4 OF 6 · TRAINING

A first-year PhD student can now get working analysis code from an agent in minutes. In their first year, they should...

A Use agents freely: directing them is the skill that matters now

B Write their core analyses by hand first, agents later

C Use agents, but explain every line at lab meeting

D Keep agents away from thesis-critical analyses

QUESTION 5 OF 6 · THE LITERATURE

Soon an agent can re-run any published analysis from its repository in minutes. What happens to your field's literature?

A Replication becomes routine and the literature gets cleaner

B A wave of corrections and retractions, then a cleaner literature

C Little changes: nobody is rewarded for checking

D AI-generated papers multiply faster than anyone can check them

QUESTION 6 OF 6 · A REALITY CHECK

In a 2025 randomised trial, experienced open-source developers worked on their own projects with and without AI tools. With AI, they were…

A about 40% faster

B about 20% faster

C no faster and no slower

D about 19% slower

METR (Model Evaluation & Threat Research), "Measuring the impact of early-2025 AI on experienced open-source developer productivity", July 2025 · 16 developers, 246 real tasks.

TONIGHT · THREE OBJECTIVES

By 19:00 you should…

1 · IMAGINE

See your place in agentic science

Your contribution to science, your career and your purpose, once agents write much of the code. What becomes yours to do?

2 · UNDERSTAND

Know how vibe coding works

Think fast and slow: let the agent go fast, check slowly. Try, fail and retry cheaply. GitHub as the safety net. Working as a team, with agents in the loop.

3 · PREPARE

Know how it runs at the hack

Whether you already have the tools or not, both paths are covered: your own setup, or the HPE / NVIDIA environment, open to every team to build and test on.

Objective 2 follows the course *Vibe Coding for Scientists* (github.com/wazaahhh/vibe-coding-crash-course): modules 01–04 were this week's pre-work; tonight draws on modules 02, 03, 05, 06 and 08.

TONIGHT · 17:00 → 19:00

Hold your answers: each one comes back tonight

TIME	BLOCK	OBJ.	QUIZ
17:00	Opening question and quiz	1	—
17:15	**Live build**: a sleep and mood analysis on phenotyping data, from question to pull request	2	Q1 Q6
17:40	**Hands-on in pairs** (breakout rooms), in a challenge you voted for: explain, plan, commit, pull request	2	Q4
18:15	**Review hunt**: find the planted errors in an AI-written analysis	2	Q2 Q6
18:35	**How the hack runs**: with or without your own tools, the HPE / NVIDIA environment, data rules	3	Q5
18:50	**Your place**: the room's answers, voices from the field, back to the opening question	1	Q1 Q3

25 min

17:15 · OBJECTIVE 2 · UNDERSTAND

Live build: from a question to a pull request

A real analysis on digital-phenotyping data, directed live, with an agent in the editor.

Your job while you watch: type in the chat whenever you see a pillar being skipped.

THE PRE-WORK IN ONE SLIDE · MODULE 02

You direct; the agent writes. Four disciplines keep it honest.

Diverge

Generation is nearly free: ask for three approaches with trade-offs, then choose.

Think critically

Every output is a pull request from a brilliant, overconfident intern. *How would I know if this is wrong?*

Fast and slow

The agent is System 1: fast, fluent, sometimes wrong. You supply System 2. The more it matters, the slower you read.

Try and fail

Attempts cost minutes. Branch, try, throw away. Git makes failure reversible.

> Trust the vibe to **write** it; never trust the vibe that it's **right**.

THE DEMO

Does last night's sleep predict today's mood?

40 participants · 21 nights, 12 Oct → 1 Nov 2026 · phone sleep detection, wearable steps, a four-item mood questionnaire every evening.

Synthetic, and messy on purpose, like the plant dataset in the pre-work: the traps teach the mindset.

Repository: `editions/facmed-hack-2026/demo-phenotyping/` in the course repo. Fork it and follow along, or just watch.

On 6 Nov the same move starts from your challenge page: a need, a breakthrough-app opportunity, one testable question.

Watch for where it goes fast, and where we slow down

#	MOVE	WHAT WE TYPE	PILLAR
1	Orient by asking	*"Brief me on `data/` and the codebook. What problems do you see?"*	Fast
2	Plan before code	*"Plan how you'd relate sleep to next-day mood. Don't write code yet."*	Diverge
3	Let it run	sleep per night → mean per night → a plot	Fast
4	Slow down at the picture	*one night looks strange: why?*	**Slow**
5	Fix with intent	*"Localise to Europe/Zurich, convert to UTC, then subtract. Show me before and after."*	Critical
6	Review and commit	`git diff` → commit → a line in `NOTES.md` → pull request	Try and fail

MODULE 06 · THE CONTROLS

Three dials, whichever agent you use

1 · Edit mode: how much rope

Plan first for anything big · **edit** with every diff approved for results that matter · **agent / auto** only on a throwaway branch, because git makes it reversible.

2 · Model: the right brain

A frontier or "thinking" model for statistics, study design and subtle bugs; a fast one for boilerplate. Escalate when it is stuck.

3 · Steering: hand on the wheel

Interrupt the moment it drifts · correct with specifics · paste the traceback · small steps · clear a thread that goes in circles.

Same dials in Claude Code, Codex, GitHub Copilot and Cline. At the hack: your own agent, or Cline on the open models served on the HPE / NVIDIA environment.

The vibe loop is the git loop

- `git diff` is how you review the agent's work: the diff is the intern's pull request.
- **Commit before a big prompt**: a guaranteed way back.
- **Branches** make the risky attempt cheap. `main` stays known-good.
- The agent can run git and `gh` for you. **You read the diff; you merge.**

WHAT JUST HAPPENED

A clean run with a wrong number

5.9 h
mean sleep on the night of 24 → 25 October, subtracting clock times

6.9 h
the same night in UTC, and every other night

- No error, no warning: the clocks went back an hour at 03:00 on Sunday.
- The agent was **mostly right**, the dangerous kind of wrong.
- Found by a **plot** and the **calendar**: domain knowledge.
- Recorded in `NOTES.md`: *verified: DST night, by plot* · *trusted: pandas time-zone handling*.

> Quiz Q6: it felt fast. The question is whether it was right.

35 min

17:40 · OBJECTIVE 2 · UNDERSTAND, BY DOING IT

Hands-on in pairs

Zoom breakout rooms of two, in the repository of a challenge one of you voted for.

Goal by 18:15: one small change, reviewed, committed, in a pull request with a note on what you checked.

MODULE 08 · PAIR ON THE PROMPTS

Two roles, switch at 17:57

Driver

Shares their screen and talks to the agent. Says out loud what they are about to ask, and why.

Skeptic

Never touches the keyboard. After every answer: *how would we know if this is wrong?* Decides when to slow down.

YOUR SITUATION	WHAT TO DO
One of you has an agent installed	That person drives first
Neither has an agent yet	Drive in **chat mode** (module 03): paste the README and code into any AI chat, never data
No access to a challenge repository yet	Use the demo: `editions/facmed-hack-2026/demo-phenotyping/`

FOUR QUESTS · DO THEM IN ORDER

From "what is this?" to a pull request

#	QUEST	THE MOVE	~MIN
1	**Explain**	Ask the agent to brief you on the repository and the challenge brief. What is missing from its answer?	5
2	**Plan**	Pick one need or breakthrough-app opportunity from the challenge page. Ask for a plan for a first, tiny analysis or prototype. No code yet. Approve or correct it	10
3	**Change**	On a branch (`firstname/topic`), let the agent make the first small step. Run it. Read the `git diff` together. Commit	12
4	**Pull request**	Push, open a pull request, and add two lines to `NOTES.md` : *verified:* … and *trusted:* …	5

At 18:15, back in the main room: paste your pull-request link (or your furthest quest) in the chat.

Five prompts worth stealing

1. *"Explain what this code does before you change anything."*
2. *"Outline your plan, then wait for my OK."*
3. *"Write a test that would fail if this result were wrong."*
4. *"What could make this conclusion wrong? Argue against yourself."*
5. *"Summarise in plain language what you changed and what you assumed."*

Give it context

The goal, the data's structure, the constraints, what you tried. Vague in, generic out.

Data stays out

The agent may see column names and made-up rows. It never sees records from a challenge dataset.

20 min

18:15 · OBJECTIVE 2 · THE SLOW PART

Review hunt

An agent wrote this analysis. It runs, it reads well, and it reaches a confident conclusion.

Would you put your name on it?

15 MIN IN YOUR PAIRS · THEN POST YOUR LIST IN THE CHAT

List everything you would not sign

1. Open `demo-phenotyping/analysis.py` and the codebook `data/README.md`.
2. Run it (or read the output on the right).
3. Hunt: the code, the data, the method, the summary.
4. For each problem: **what tipped you off?**

Allowed: plots, the codebook, a second AI, your own judgment. The prompt the agent got is at the top of the file.

Strongest on syntax, weakest on truth

TRUST, BUT SKIM	SLOW DOWN: IT FAILS QUIETLY HERE
Plotting, formatting, refactoring	**Domain correctness**: units, time zones, reverse-keyed items
Boilerplate, glue, file reading	**Statistics**: the wrong-but-plausible test, leakage
Explaining unfamiliar code	**What the outcome really measures**
Translating between languages	**Anything needing ground truth**: codebooks, citations

> The bug that ends a paper compiles, runs, and looks beautiful.

THE ANSWER KEY

Five problems, and the headline flips

#	PROBLEM	EFFECT	WHAT CATCHES IT
1	`mood_3` reverse-keyed, averaged as is	mood spread halves; r(sleep) +0.45 → +0.39	the codebook
2	Non-wear days kept as 0 steps	fake steps–mood link: r +0.15 → +0.28	a histogram: spike at 0
3	Same participants in train and test	AUC 0.89 → **0.69** on new people	"would it work on a new patient?"
4	Summary contradicts its numbers	"activity is stronger": 0.25 < 0.39	numbers before prose
5	Overclaim	a low-mood *day* is no depressive *episode*	what does the outcome measure?

Fixed: sleep predicts next-day mood (r ≈ +0.45), steps add little (r ≈ +0.15), the classifier generalises modestly (AUC ≈ 0.69).

MODULE 08 · WORKING TOGETHER

When code is cheap, review is the work

Peer review matters **more** when generation is cheap: the failure mode is unreviewed AI code flowing straight into results.

Show how you checked

Demo what you built and how you verified it.

A second pair of eyes

No AI-written analysis reaches a claim without a human reading the reasoning.

Keep the record

Whose prompt, which model, what was verified and what was trusted.

15 min

18:35 · OBJECTIVE 3 · PREPARE

How it runs at the hack

Friday 6 – Saturday 7 November · Campus Biotech · 32 hours.

WITH OR WITHOUT YOUR OWN TOOLS

Two ways in, one team repository

You already have the tools

VS Code on your laptop with your own agent: Claude Code or Codex on your own plan, GitHub Copilot, or Cline. Clone your team's repository and work as tonight.

You don't, or you'd rather not install

A ready-made VS Code in the browser on the HPE / NVIDIA environment: nothing to install, an agent already set up on open models served there, your team's repository already cloned.

> **Either way, every team can build and test on the HPE / NVIDIA environment**: GPUs for heavy jobs, and the challenge datasets, which stay there.

THE WORKFLOW YOU REHEARSED TONIGHT

Your team's repository

RULE	WHY
One repository per team, in the hack's GitHub organisation	One place for code, notes and the demo
A branch per person and task: `firstname/topic`	`main` stays working while you try things
Commit small and often, under your own name	For credit candidates, the contribution history is the assessment
Merge through a pull request, with `NOTES.md` updated: *verified* and *trusted*	The record of what was checked, by whom
Tag `demo` on `main` before the final presentations	The jury looks at exactly what you showed

Each repository comes with an `AGENTS.md`: the same rules, written for your agent, so it branches, commits and opens pull requests the way the team does.

DATA RULES · THE SHORT VERSION OF CRASH COURSE #1

The agent sees the structure of the data, never the records

Real data stays put

Challenge datasets live on the HPE / NVIDIA environment, read-only. Analyses run there.

Laptops get samples

Each repository has a small synthetic sample in `examples/`, so code runs anywhere.

Nothing sensitive in git

A pre-commit check blocks data files and secrets. Never paste records into a chat tool.

BEFORE 6 NOVEMBER

Five things to do

	WHAT	WHEN
1	**Vote** on the challenges you want to work on	Now: voting is open
2	A **GitHub account** with two-factor authentication, and your username sent to us	This week
3	*Optional:* VS Code + one agent on your laptop; try the demo repository	Before 6 Nov
4	**Crash Course #3**: grant writing for pop-up projects, with the RGO	Thu 29 Oct, 17:00
5	Come with one idea for **your** challenge: a need, an opportunity, a question	6 Nov, 11:00

10 min

18:50 · OBJECTIVE 1 · IMAGINE

Your place in agentic science

Back to where we started: what you imagined at 17:00, and what you saw tonight.

THE ROOM'S ANSWERS

What you imagined, and voices to argue with

Q	THE ROOM	A VOICE FROM THE FIELD
1 · Where first	*poll*	Agentic coding is already removing the engineering bottleneck in modelling (DePasquale)
2 · Scarce skill	*poll*	LLMs are *"a flashlight as opposed to a laser pointer"* (Kim Stachenfeld, DeepMind)
3 · Depth	*poll*	*"Prolific but shallow—generating models faster than we can generate insights"* (DePasquale)
4 · Training	*poll*	Struggling to turn an idea into code is *"a form of disciplined thinking"* (DePasquale)
5 · Literature	*poll*	Incentives already reward producing models over understanding them; vibe modelling could make it worse (Dan Goodman, Imperial)
6 · Reality	**D** · 19% slower	…while feeling 20% faster (METR, 2025)

THE OPENING QUESTION, REVISITED

Where will agentic coding change science?

What we saw tonight

Analysis and modelling in minutes · a plan before the code · review as the real work · a pull request as the record of what was checked

Still imagination

Your answers from 17:00 that nothing tonight showed yet: where your own research, career or field could go

> **Last line in the chat:** *"When agents write the code, the part of my research that stays mine is …"*

SEE YOU ON 6 NOVEMBER

Director, not author.

The agent brings **speed and breadth**. You bring **the question, the judgment and the responsibility**. That division of labour is the skill.

The course

github.com/wazaahhh/vibe-coding-crash-course · fork it, star it, open a pull request

Read

DePasquale, *Models at the speed of thought* · Schuck, *How, when and why to use agentic AI in our neuroscience labs* · both in The Transmitter, 2026

Next

Vote on challenges · GitHub account · Crash Course #3 on Thu 29 Oct

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